

Appendix

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A. Formal Boundary

The formal core models a deterministic straight-line template over a semantic value (b, a, d) and an accumulator y. A read transition exposes f(b, a, d), increments access count, and may update latent drift. An observation transition exposes no additive value and updates only latent drift through g. Body execution is a deterministic fold over the operation list, followed by a deterministic cap.

The proved claims are fixed-order determinism, zero divergence for identity observations, zero divergence for access-insensitive reads, and preservation of body-level divergence by a nonzero-slope linear cap. The proof core does not claim universal nonlinear amplification, analyzer soundness, arbitrary Python soundness, or production prevalence.

B. Real-Code Oracle Summary

The real-code oracle selected 60 candidates and constructed 39 executable harnesses. It confirmed 20 divergences across 12 packages. The confirmed divergences comprise 1 branch divergence, 17 output divergences, and 2 state-only divergences. Nine caller-level wrappers converted confirmed divergences into downstream branch consequences. Nineteen negative controls removed the divergence under fresh-object, reset-between, or pure-observation interventions.

C. Primary Benchmark Matrix

Model	Tasks	Executable	Preserved	Ordinary bugs	Invalid patches	Verified OSDS	Silent OSDS
gpt-5.6-sol	26	26	23	3	0	0	0
gpt-5.6-terra	26	24	18	4	2	2	2
gpt-5.6-luna	26	24	17	4	2	3	3

Verified OSDS rows:

Model	Task	Mechanism
gpt-5.6-terra	pytest_catching_logs__instrumentation__normal	handler level changed by logging-shaped access
gpt-5.6-terra	pytest_catching_logs__instrumentation__warned	handler level changed by logging-shaped access
gpt-5.6-luna	pytest_catching_logs__instrumentation__normal	handler level changed by logging-shaped access
gpt-5.6-luna	pytest_catching_logs__instrumentation__warned	handler level changed by logging-shaped access
gpt-5.6-luna	pyyaml_representer__caching_materialization__normal	identity/access cache state reused after observation

D. Causal-Control Matrix

Failure family	Rows	Original replay	Controlled replay	Causal status
pytest catching_logs instrumentation	4	ordinary-pass, OSDS-fail	OSDS-pass	mechanism_neutralized_divergence_disappeared
PyYAML representer caching	1	ordinary-pass, OSDS-fail	OSDS-pass	mechanism_neutralized_divergence_disappeared

All five exact generated transformations were preserved byte-identically during the control experiment.

E. Prospective Expansion Matrix

Model	Tasks	Executable	Preserved	Ordinary bugs	Invalid patches	Verified OSDS	Silent OSDS
gpt-5.6-sol	14	14	14	0	0	0	0
gpt-5.6-terra	14	14	14	0	0	0	0
gpt-5.6-luna	14	14	14	0	0	0	0

Expansion witnesses:

Witness	Package	Family
re07_boltions_LRI	boltions	repeated_access_cleanup
re09_boltions_MultiFileReader	boltions	access_reordering
re13_h11_ReceiveBuffer	h11	access_reordering
bs15_boltions_SpooledStringIO	boltions	access_reordering
ext02_boltions_SpooledBytesIO	boltions	access_reordering
ext07_dnspython_Tokenizer_concat	dnspython	access_reordering
ext08_boltions_LRU_pair2	boltions	repeated_access_cleanup

All expansion rows are expected-access-sensitive. Hidden-observation rows are present in the primary benchmark only.

F. Reproducibility Pointers

Primary benchmark tasks are under `experiments/agent_behavior_preservation/benchmark/tasks.jsonl`. The prospective expansion tasks are under `benchmark_expansion/tasks.jsonl`, with frozen prompts in `benchmark_expansion/prompts/`. Raw Codex task-model expansion responses are saved under `benchmark_expansion/responses/`. Replays use `experiments/agent_behavior_preservation/runners/run_benchmark.py --provider jsonl --task-ids-from-replay`.

The causal controls are under `experiments/agent_behavior_preservation/causal_controls/` and produce `analysis/model_failure_causal_controls.csv` and `.md`. The expansion summary artifacts are `analysis/prospective_expansion_results.csv`, `analysis/prospective_expansion_results.md`, and `analysis/model_differential_cases.md`.